

Interactive Demonstration of Active Bed-Frame Vibration Sensing for Respiratory Airflow Monitoring

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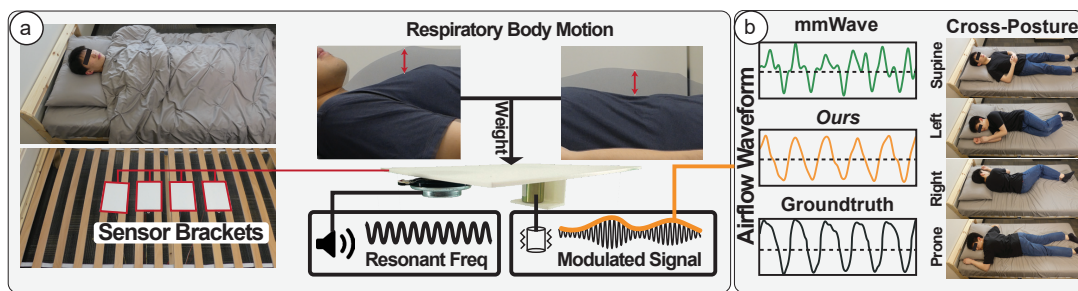


Figure 1: (a) Our system reconstructs an airflow-correlated respiratory waveform through active bed-frame vibration sensing. Respiration cyclically redistributes body weight, modulating the bed’s structural vibration response. A co-designed sensor bracket and a resonance-aligned excitation improve sensitivity while remaining imperceptible. (b) Across diverse sleep postures, our system reconstructs this waveform more robustly than mmWave sensing.

Abstract

Respiratory airflow waveform monitoring is a crucial physiological signal for sleep disorders and critical care monitoring. However, current clinical approaches often require cumbersome instrumentation, which limits their practicality for everyday use. We present a non-invasive system that reconstructs an airflow-correlated respiratory waveform across unconstrained sleep postures. Our system injects an imperceptible single-tone vibration into an ordinary bed frame and measures how respiration-induced body-load redistribution modulates the frame’s structural vibration response. A co-designed L-shaped sensor bracket and resonance-aligned excitation enhance sensitivity to these subtle respiratory effects, while a real-time signal-processing pipeline demodulates the measured response into a continuous respiratory waveform.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; *Ubiquitous and mobile computing systems and tools*; • **Applied computing** → *Consumer health*.

Keywords

Active Vibration Sensing; Sleep Monitoring; Respiratory Airflow; Structure-borne Sensing; Unobtrusive Health Sensing

ACM Reference Format:

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1 Introduction

Respiratory airflow is a primary clinical signal for identifying sleep-related breathing events because its waveform captures the timing and relative magnitude of each breath [6]. However, clinical airflow measurement typically requires nasal cannulas and multi-channel polysomnography, limiting its practicality for nightly in-home use. Existing wearables often provide only respiratory rate [7], while contactless approaches based on radar, WiFi, or audio primarily recover respiratory effort from local thoracoabdominal motion and can be sensitive to posture and occlusion [8, 9].

We present an active bed-frame vibration sensing system that reconstructs an airflow-correlated respiratory waveform across unconstrained sleep postures. The key insight is that breathing periodically redistributes body load through diaphragmatic and



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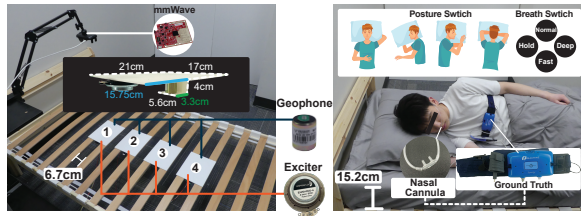


Figure 2: The demonstration setup. Four active sensing modules are mounted on a bed. A co-located mmWave radar serves as the baseline, while an airflow sensor is the ground truth.

thoracoabdominal motion, thereby modulating the structural vibration response of the bed [4]. An imperceptible vibration is injected as an interrogation signal and demodulates respiration-induced changes in the measured response into a continuous respiratory waveform. As these respiratory modulations are weak, our system combines resonance-aligned excitation with a compliant sensor bracket to improve sensitivity while keeping the excitation imperceptible.

2 System Overview

Respiration-induced weight redistribution. During breathing, diaphragmatic motion and thoracoabdominal expansion redistribute body load across the bed. These whole-body load shifts follow respiratory timing and relative breath intensity, but their effects on the bed frame are too weak to measure reliably through passive vibration sensing alone.

Active vibration sensing. Our system leverages an imperceptible single-frequency vibration to excite the bed frame and measures how respiration-induced load changes modulate the structural response. A compliant L-shaped bracket co-locates the sound exciter and geophone, while resonance-aligned excitation increases sensitivity to the weak respiratory modulation without requiring perceptible vibration levels.

Airflow-correlated waveform extraction. Our system isolates the excitation frequency, extracts its amplitude envelope using the Hilbert transform, and low-pass filters the envelope to recover a continuous airflow-correlated respiratory waveform. The system automatically selects the highest-quality sensing channel and displays the reconstructed waveform and respiratory rate in real time.

3 Implementation and Results

As shown in Figure 2, our system consists of four active sensing modules mounted beneath a wooden bed frame and distributed from the head to the hip region. Each module pairs a Dayton Audio EX25HRDS-4 sound exciter [1] with an SM-24 geophone [5]. The geophone signal is amplified, filtered, and digitized using an Analog Discovery Pro [2]. A one-time frequency sweep identifies the dominant resonance of the coupled exciter–bracket–frame structure. The bedframe operates using a 55 Hz single-tone excitation. Before each demonstration session, the excitation amplitude is briefly calibrated to remain below the visitor’s audible and tactile perception thresholds.

We implement a real-time streaming pipeline that continuously reconstructs the airflow-correlated respiratory waveform with sub-second latency. Respiratory rate and breathing-pattern predictions are updated over rolling windows and displayed on a live dashboard. To provide direct on-site references, the setup also includes a co-located mmWave radar positioned above the torso and an optional lightweight respiratory-effort belt. The belt provides an intuitive visual reference for respiratory timing and relative depth, while airflow-level validation is based on measurements collected during the offline user study.

We evaluated our system with 21 healthy participants using a nasal cannula from an FDA-cleared home sleep monitor as the airflow reference [3]. Across common sleep postures, our system achieved a mean waveform correlation of 0.71, compared with 0.50 for mmWave. It also reduced respiratory-rate error from 4.41 to 1.88 breaths per minute. For unseen participants, breathing-pattern classification achieved 81.2% accuracy, demonstrating the system’s ability to generalize to new users.

4 Conclusion

We present an active vibration sensing system that reconstructs an airflow-correlated respiratory waveform from imperceptible bed-frame vibrations. By combining respiration-induced load redistribution with a compliant sensor bracket and resonance-aligned excitation, our system recovers respiratory dynamics across different sleep postures and outperforms a mmWave baseline in waveform reconstruction and respiratory-rate estimation.

References

- [1] Dayton Audio. 2026. EX25HRDS-4 IMS™ High Roll Dual Suspension 25mm Interchangeable Hardware Mount Exciter 20W 4 Ohm. <https://www.daytonaudio.com/product/1957/ex25hrds-4-ims-high-roll-dual-suspension-25mm-interchangeable-hardware-mount-exciter-20w-4-ohm>. Accessed: 2026-03-31.
- [2] Digilent. 2026. Analog Discovery Pro (ADP3450/ADP3250). <https://digilent.com/reference/test-and-measurement/analog-discovery-pro-3x50/start>. Accessed: 2026-03-31.
- [3] General Sleep Corporation. 2026. Zmachine® Synergy. <https://www.generalsleep.com/zmachine-synergy.html>. Accessed: 2026-03-31.
- [4] Hewon Jung, Jacob P Kimball, Timothy Receveur, Asim H Gazi, Eric D Agdeppa, and Omer T Inan. 2022. Estimation of tidal volume using load cells on a hospital bed. *IEEE Journal of Biomedical and Health Informatics* 26, 7 (2022), 3330–3341.
- [5] Sparkfun. 2006. SM-24 Geophone Element. <https://cdn.sparkfun.com/datasheets/Sensors/Accelerometers/SM-24%20Brochure.pdf>. Accessed: 03-20-2024.
- [6] Patrick J Strollo Jr and Robert M Rogers. 1996. Obstructive sleep apnea. *New England Journal of Medicine* 334, 2 (1996), 99–104.
- [7] Xiao Sun, Li Qiu, Yibo Wu, Yeming Tang, and Guohong Cao. 2017. Sleepmonitor: Monitoring respiratory rate and body position during sleep using smartwatch. *Proceedings of the ACM on interactive, mobile, wearable and ubiquitous technologies* 1, 3 (2017), 1–22.
- [8] Tianben Wang, Daqing Zhang, Leye Wang, Yuanqing Zheng, Tao Gu, Bernadette Dorizzi, and Xingshe Zhou. 2018. Contactless respiration monitoring using ultrasound signal with off-the-shelf audio devices. *IEEE Internet of Things Journal* 6, 2 (2018), 2959–2973.
- [9] Zhicheng Yang, Parth H Pathak, Yunze Zeng, Xixi Liran, and Prasant Mohapatra. 2016. Monitoring vital signs using millimeter wave. In *Proceedings of the 17th ACM international symposium on mobile ad hoc networking and computing*. 211–220.